

PROJECT REVIEW

Superfecta Project Report

Final report based on the completed full training reruns. This version reflects the successful end-to-end results on both the major-tracks dataset and the full all-tracks dataset, not the earlier failed run.

Updated Apr 13, 2026

Harville + XGBoost Residual Model

Full Training Reruns Complete

Executive Summary

The current Harville + XGBoost residual model is a strong payout-estimation baseline, but it is not yet a complete profitable betting system. The full reruns confirm that the model consistently improves plain Harville when predicting realized superfecta payout distortion.

Main conclusion: the current model is validated as a useful payout-correction baseline, but the best next step is a two-stage expected-value system that combines hit probability, payout prediction, and field-size-specific ticket selection.

What the Current Project Does

Current Modeling Logic

- Uses Harville probabilities as the baseline for superfecta combinations
- Trains XGBoost on **log(actual payout / Harville payout)**
- Applies the model to historical prediction and live NYRA scoring
- Targets payout distortion, not direct combo-hit probability

Assessment

- The idea is legitimate and the data has real signal
- The payout model works much better than plain Harville
- Profitability is still not directly modeled
- A true ROI engine still needs ranking, EV filtering, and ticket selection

Full Training Rerun Results

Two full reruns completed successfully after fixing the zero-total-probability preprocessing bug.

DATASET	RACE- LEVEL EXAMPLES	FEATURES	LOG- RATIO TEST RMSE	HARVILLE RMSE	ML- CORRECTED RMSE	RMSE IMPROVEMENT	MAE IMPROVEMENT
14years_major_tracks.csv	88,974	52	0.3757	\$46,485.05	\$8,921.57	80.8%	77.5%
14years.csv	552,926	55	0.4534	\$21,875.34	\$4,718.26	78.4%	75.5%

BEST TARGET FIT

0.3757

Major-tracks log-ratio test RMSE

BEST PAYOUT RMSE

\$4.7k

All-tracks ML-corrected RMSE

TOP RMSE GAIN

80.8%

Major-tracks vs Harville

TOP MAE GAIN

77.5%

Major-tracks vs Harville

Bottom line from the reruns: both datasets confirm that the residual model is capturing meaningful structure in superfecta payout distortion and materially outperforming plain Harville for payout estimation.

How to Read These Results

What the results do prove

- The modeling direction is valid
- XGBoost is learning real payout-distortion signal
- Harville is a beatable baseline for payout estimation
- The major-tracks and all-tracks datasets both support this

What the results do not prove

- They do not directly prove profitable betting
- They do not model exact-combo hit probability yet
- They do not yet include a bankroll or ticket strategy layer
- They do not replace the need for a walk-forward ROI backtest

Important caution: this is a payout-correction win, not final proof of profitable wagering. Profitability still depends on ranking the right tickets, filtering by expected value, and structuring bets correctly by field size.

Best Path Forward

The best near-term strategy is a two-stage expected-value framework.

1. **Stage 1:** predict the probability that a specific superfecta combination wins

2. **Stage 2:** predict payout distortion or expected payout for that combination
3. **Decision layer:** bet only combinations whose expected value exceeds takeout and risk thresholds

Recommended Field-Size Bins

- 6-7 runners
- 8-9 runners
- 10-11 runners
- 12+ runners

Recommended Next Steps

1. Build a walk-forward backtest harness based on race date, not just random train-test splitting
2. Keep the current residual model as the payout-estimation baseline
3. Build an exact-combination ranking model for hit probability
4. Combine the ranker and payout model into a two-stage EV engine
5. Add field-size-specific no-bet thresholds and ticket budget rules
6. Later, expand to richer live-market features such as odds drift and late probables

Engineering Notes

- Preprocessing was fixed so races with zero total implied probability are skipped instead of crashing
- Training scripts were patched to support explicit CPU thread usage
- Prediction scripts were patched to support threads and to load the XGBoost model once instead of inside the inner loop

Final Answer

The final reruns confirm that the current XGBoost residual model meaningfully improves Harville as a payout estimator on both the major-tracks and all-tracks datasets. The correct next step is not to stop here, but to build a two-stage EV system that combines exact-combination hit probability, payout prediction, and field-size-specific ticket selection.

Generated from the completed full training reruns in Superfecta Help.